INCH-POUND
MIL-P-70875 (AR)
10 March 1992

MILITARY SPECIFICATION

PRINTED WIRING BOARD ASSEMBLY, TRIGGER INTERFACE (M2/M3), SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the Printed Wiring Board Assembly (PWBA), Trigger Interface (M2/M3), part number 9352750.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1265

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	-	Environments, Working, Minimum Standards for
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	-	Calibration System's Requirements

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HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

9352750 - Printed Wiring Board Assy, Trigger Interface

9352752 - Schematic Diagram, Trigger Interface

PACKAGING DATA SHEET (CAGE CODE 19200)

SPI-P9352750 - Printed Wiring Board Assy, Trigger Interface

(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. The Trigger Interface board is located in the Loaders Control Assembly. The Trigger Interface Assembly contains a voltage regulator to power the logic circuit before isolation, logic circuits and four optical isolators which serve to decouple the vehicle power from the remainder of the MILES system.

Trigger closures by the operator produce logic output signals from the board which identify the weapon being fired and thereby initiate the appropriate MILES response. The logic circuitry incorporated permits proper performance with a wide range of different vehicle trigger logic systems.

3.2 First article. When specified (see 6.4), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental stress screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°C and +62.8°C. After exposure to ESS in accordance with 4.5, the characteristics of 3.4 shall not be degraded.

3.4 Characteristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table I, measured under quiescent conditions. For terminal and component designations called out in this specification, see drawing 9352752, Schematic Diagram, Trigger Interface.

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TABLE I. Operating Conditions.

Parameter	Terminal	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-31.7	25	62.8	°C
Input Power (+) V_{TR}	2	18	24	31	Vdc
(+) V_D	63, 64	4	4.5	6	Vdc
(+) V_B	71, 72	7	9	12	Vdc
Return (-)	1, 3, 4				
Power (-)	67, 68				
(-)	67, 68				
Input Current V_{TR}	2		10		mA
V_D	63, 64		2		mA
V_B	71, 72		5		mA

3.4.2 Performance. Unless otherwise specified, the PWBA shall meet the requirements of this specification under the following conditions:

a.	Temperature	25° ± 10°C (77° ± 18°F)
b.	Relative humidity	50 ± 30 %
c.	Barometric pressure	725 + 50, -75 mm Hg (28.5 + 2.0, -3.0 inches Hg)
d.	Input power and loads	See Figure 1

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testability. Each LRU shall contain test points in accordance with 3.4.1

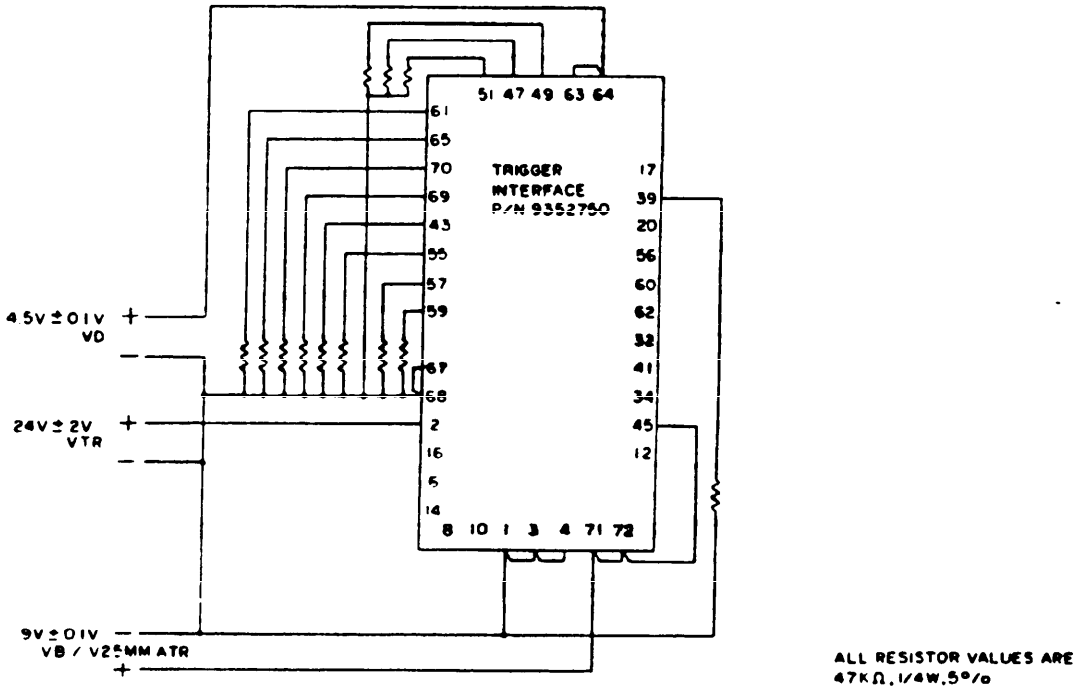


FIGURE 1. Trigger Interface PWBA Power Source and Loads.

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3.4.2.1 Current drain. With the voltage sources V_{TR}, V_D and V_B applied, the current drain of the board shall be:

Terminal	Current Drain (mA)
2	10 (nominal)
63	2 (nominal)
71	5 (nominal)

3.4.2.2 Stable state. The voltages at six terminals shown below define the stable state of the board. With power applied, the output voltages shall be:

Terminal	Voltage (Vdc)
17	5 ± 0.5
39	0 (+0.01,-0.01)
20,56,60,62	4.5 (+0.1,-0.6)

3.4.2.3 Wire cut. The voltages at five terminals shown below define the wire cut function. With power applied, and terminals 2 and 16 (TOW latch) connected, optical isolator U4 turns on and terminal 20 (wire cut) will go low to identify the presence of TOW latch. The output voltages shall be:

Terminal	Voltage (Vdc)
20	0 (+0.5,-0)
39	0 (+0.01,-0.01)
56,60,62	4.5 (+0.1,-0.6)

3.4.2.4 TOW trigger. The voltages at five terminals shown below define the TOW trigger state for the board. With power applied, terminals 2 and 16 (TOW latch) connected, and terminals 1 and 8 (missile 1 present) connected, the output voltages shall be as shown below. With terminals 1 and 8 disconnected and terminals 1 and 10 (missile 2 present) connected, the same results shall occur. When TOW latch and missile are present, optical isolator U5 turns on and terminal 60 (TOW missile) goes low. With a missile present, pin 1 of D-latch U3 goes high and starts charging capacitor C7 through resistor R9. After 130ms, chip U3 is reset, isolator U5 is turned off and terminal 60 goes high.

Terminal	Voltage (Vdc)
20	0 (+0.5,-0)
39	0 (+0.01,-0.01)
56,62	4.5 (+0.1,-0.6)
60	(see Figure 2)

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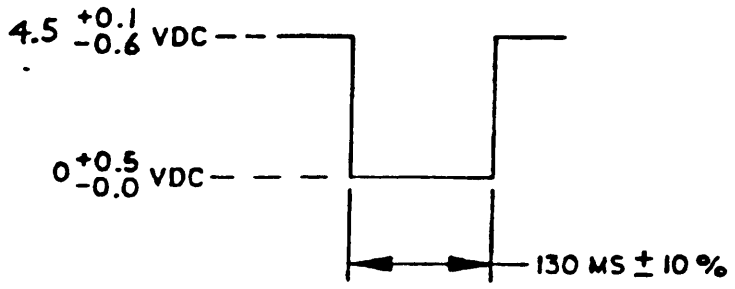


FIGURE 2. TOW Trigger.

3.4.2.5 COAX trigger. The voltages at five terminals shown below define the COAX trigger state for the board. With power applied and terminals 2 and 6 (actual COAX trigger) connected, the output voltages shall be as shown below. Terminal 56 (COAX trigger) goes low when the optical isolator U6 is activated by the presence of the actual COAX trigger.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
56	0 (+0.5,-0)
20,60,62	4.5 (+0.1,-0.6)

3.4.2.6 25 mm trigger. The voltages at five terminals shown below define the 25mm trigger state for the board. With power applied, optical isolator U7 is activated when terminals 2 and 14 (weapon) are connected and terminals 12 (actual 25mm trigger) and 71 are connected. The 25mm gun trigger (terminal 62) is enabled low with U7 activated and the output voltages shall be as shown below. Relay K1 shall not be activated by the presence of a rounds count since the AWESS signal (terminal 32) is not present. With terminals 61, 65, 70, 69, 54, 55, 57 or 59 and 63 connected, and with terminals 51 and 47 (momentarily) connected to terminal 63, the rounds count signal is present and flip flop U12 is set.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
62	0 (+0.5,-0)
20,56,60	4.5 (+0.1,-0.6)

3.4.2.7 Weapons disable. The voltages at five terminals shown below will define the weapons disable state for the board. With power applied, terminals 51 (WRT) and 61 (DB0) connected to

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terminal 63, terminals 12 (25 mm actual trigger) and 71 connected, and terminals 47 (N1) and 63 momentarily connected, the output voltages shall be as shown below. The 25mm gun trigger (terminal 62) is high since no weapon signal (terminal 14) is present to allow the U7 optical isolator to turn on. As in 3.4.2.6, the presence of a rounds count does not activate relay K1.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
20,56,60,62	4.5 (+0.1,-0.6)

3.4.2.8 25 mm trigger disable. The voltages at five terminals shown below will define the 25mm trigger disable state for the board. With power applied, terminals 51 (WRT) and 63 connected, terminals 2 and 14 (weapon) connected, terminals 61 (DB0) and 63 connected, and terminals 47 (N1) and 63 momentarily connected, the output voltages shall be as shown below. The 25mm gun trigger (terminal 62) is high since no actual 25mm trigger (terminal 12) is present to activate the U7 optical isolator. As in 3.4.2.6, the presence of a rounds count does not activate relay K1.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
20,56,60,62	4.5 (+0.1,-0.6)

3.4.2.9 Microphone trigger. The voltages at five terminals shown below will define the microphone trigger for the board. With power applied, terminals 51 (WRT) and 63 connected, terminals 2 and 14 (weapon) connected, terminals 1 and 34 (microphone) connected, terminals 61 (DB0) and 63 connected, and terminals 47 (N1) and 63 momentarily connected, the output voltages shall be as shown below. The 25mm gun trigger (terminal 62) is low since the microphone signal enabled "and" gate U10, pin 3. As in 3.4.2.6, the presence of a rounds count does not activate relay K1.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
62	0 (+0.5,-0)
20,56,60	4.5 (+0.1,-0.6)

3.4.2.10 AWESS trigger. The voltages at five terminals shown below will define the AWESS trigger for the board. With power applied, terminals 32 (AWESS/DRY), 51 (WRT) and 61 (DBO) connected to terminal 63, terminals 12 (trigger) and 71

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connected, terminals 2 and 14 (weapon) connected, and terminals 47 and 63 momentarily connected, the output voltages shall be as shown below. The AWESS trigger (terminal 39) is asserted high and 25mm gun trigger (terminal 62) is disasserted low when AWESS/DRY (terminal 32) is high. With terminal 32 high, "NAND" gate U8 at pin 10 is disabled which disasserts "AND" gate U10 at pin 3 and terminal 62 even though optical isolator U7 is enabled. Terminal 32, along with optical isolator U7, also enables "AND" gate U10 at pin 4 which enables "NAND" U9 at pin 5. This, along with the presence of a rounds count, enables relay K1 and sets terminal 39 high.

Terminal	Voltage (Vdc)
20,56,60,62	4.5 (+0.1,-0.6)
39	9 (+0.1,-0.8)

3.4.2.11 AWESS/25 mm trigger. The voltages at five terminals shown below will define the AWESS/25mm trigger for the board. With power applied, terminals 32 (AWESS), 51 (WRT) and 61 (DBO) connected to terminal 63, terminals 34 (microphone) and 1 connected, terminals 12 (actual 25m trigger) and 71 connected, terminals 2 and 14 (weapon) connected, and terminals 47 (rounds count) and 63 momentarily connected, the output voltages shall be as shown below. Adding the microphone signal to the AWESS trigger asserts the 25mm trigger low. Even though optical isolator U7 is on, "NAND" U8 at pin 10 is disasserted since terminal 32 is asserted. Terminal 62 is asserted though since terminal 34 is low. Relay K1 is turned on since a rounds count is present at flip flop U12 setting pin 2 high and terminal 32 along with the activated optical isolator U7 enables "AND" U10 at pin 4. Relay K1 is then turned on, pulling terminal 39 high, by "nand" U9 at pin 5 one end of the relay's coil at pin 6.

Terminal	Voltage (Vdc)
62	0 (+0.5,-0)
39	9 (+0.1,-0.8)
20,56,60	4.5 (+0.1,-0.6)

3.4.2.12 Trigger disable. The voltages at five terminals shown below will define the trigger disable for the board. With power applied, terminals 2 and 14 (weapon) connected, terminals 12 (actual 25mm trigger) and 71 connected, and terminals 32 (AWESS/DRY) and 49 (kill) connected to terminal 63, the output voltages shall be as shown below. The high at terminal 49 sets flip flop U12, keeping pin 2 low, and disasserts "NAND" U12 at pin 5. This prevents relay K1 from shorting the line to terminal 39 to a high voltage. Since terminal 32 is high, "NAND" U8 at

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pin 11 is low, "NAND" U8 at pin 10 is high, even though optical isolator U7 is on, and "AND" U10 at pin 3 is high.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
20,56,60,62	4.5 (+0.1,-0.6)

3.4.2.13 FlashWESS trigger. The voltages at five terminals shown below will define the flashWESS trigger for the board. With power applied, terminals 51 (WRT) and 61 (DB0) connected to terminal 63, terminals 14 (weapon) and 2 connected, terminals 12 (actual 25mm trigger) and 71 connected, terminals 41 (flashWESS) and 1 connected, and terminals 47 (rounds count) and 63 momentarily connected, the output voltages shall as shown below. When terminal 32 is low, and terminal 41 is enabled, pin 9 at gate U8 is enabled so that the 25mm actual trigger at terminal 12 can pass through to TMGUN (terminal 62). AWESS trigger is not enabled since there is no AWESS/DRY signal.

Terminal	Voltage (Vdc)
39	0 (+0.01,-0.01)
62	0 (+0.5,-0)
20,56,60	4.5 (+0.1,-0.6)

3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subject to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2. when exposed to and after the following.

3.4.3.1.1 Operating temperature. The assembly shall operate up to a temperature of +62.8°C (+145°F) after exposure at 62.8°C for 4 hours.

3.4.3.2 Low temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -31.7°C (-25°F) after exposure at -31.7°C for 4 hours.

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -34.4°C (-30°F) for a period of 24 hours.

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3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 3. Apply three shocks (18 total) in each direction of the three mutually perpendicular axes.

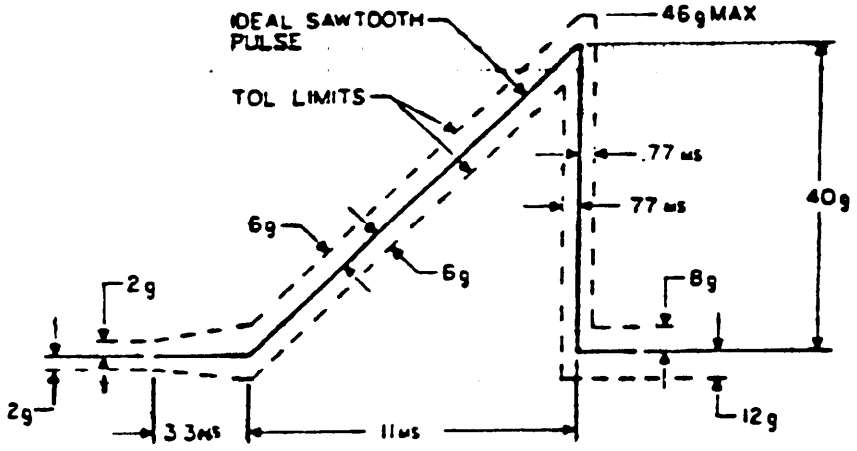


FIGURE 3. Shock Pulse.

3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 4 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz in 15 minutes
30-50 Hz	0.033" DA	
50-500 Hz	±4.2 g	

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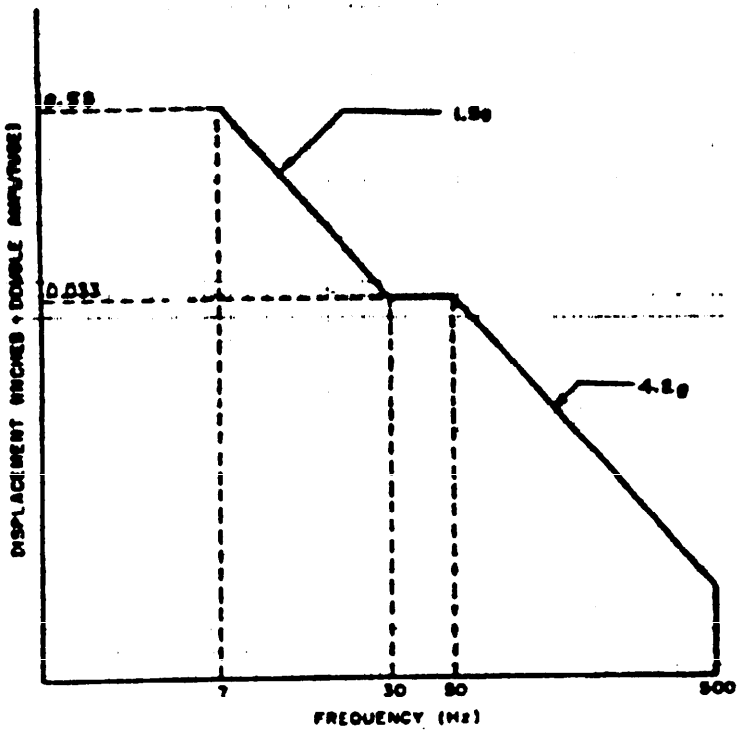


FIGURE 4. Vibration Test.

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3.5 Design and construction. The PWBA shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawing 9352750. The PWBA shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Environmental Stress Screen (see 4.5)

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. First article inspections are listed in Tables II and III. Each first article sample unit shall be inspected in accordance with Table II before being subjected to the environmental tests in Table III in the sequence listed. As determined by the Government, the first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE II. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Requirements Paragraph	Test/ inspection Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Accept Criteria
101	Current Drain	3.4.2.1	4.8.1.1	3	32/Acc-0,Rej-1
102	Stable State	3.4.2.2	4.8.1.2	3	32/Acc-0,Rej-1
103	Wire cut	3.4.2.3	4.8.1.3	3	32/Acc-0,Rej-1
104	TOW Trigger	3.4.2.4	4.8.1.4	3	32/Acc-0,Rej-1
105	COAX Trigger	3.4.2.5	4.8.1.5	3	32/Acc-0,Rej-1
106	25mm Trigger	3.4.2.6	4.8.1.6	3	32/Acc-0,Rej-1
107	Weapons Disable	3.4.2.7	4.8.1.7	3	32/Acc-0,Rej-1
108	25mm Trigger Disable	3.4.2.8	4.8.1.8	3	32/Acc-0,Rej-1
109	Microphone Trigger	3.4.2.9	4.8.1.9	3	32/Acc-0,Rej-1
110	AWESS Trigger	3.4.2.10	4.8.1.10	3	32/Acc-0,Rej-1
111	AWESS/25mm Trigger	3.4.2.11	4.8.1.11	3	32/Acc-0,Rej-1
112	Trigger Disable	3.4.2.12	4.8.1.12	3	32/Acc-0,R 1
113	FlashWESS Trigger	3.4.2.13	4.8.1.13	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Conformal Coating	3.5	Visual	3	125/Acc-0,Rej-1
204	Dim Characteristics	3.5	SME 5/	3	125/Acc-0,Rej-1
205	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
206	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 class (major characteristics) performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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TABLE III. Environmental Tests. 1/

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size, Accept Criteria
114	High Temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
115	Low Temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
116	Shock	3.4.3.3	4.9.3	3	1/
117	Vibration	3.4.3.4	4.9.4	3	1/

NOTE:
1/ Not required for quality conformance inspection.

4.4.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be subjected to a thermal stress screen in accordance with Table IV and the procedure below.

TABLE IV. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/ Group A Test
-34.4°C	+62.8°C	15°C/minute	Off	No	20/Failure Free or Additional Cycle

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4.5.1 Conformal coating. Thermal stress screening shall be performed prior to conformal coating unless another method is approved by the Government.

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -34.4°C to + 62.8°C (-30°F to + 145°F) and the duration of one cycle to be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (see Figure 5).

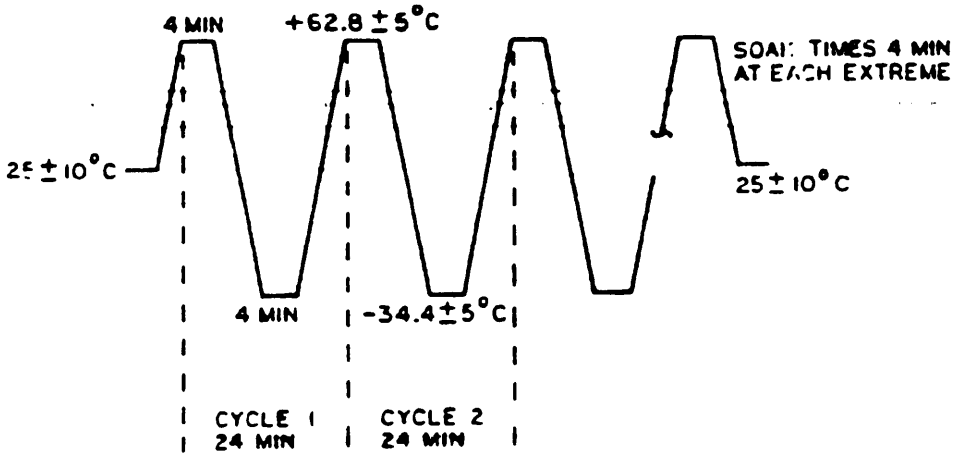


FIGURE 5. Environmental Stress Screen Temperature Profile.

4.5.3 Cycle frequency. A total of twenty temperature cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling shall be conducted with units nonoperating and without power (static).

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.6.2). If a unit is rejected, the unit shall be repaired for the failure (if possible) and be put through one additional cycle of ESS. The unit shall then undergo and pass the Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from

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which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table II in the quantities specified.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table III and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements of 5.1. During the period of first article inspection, one additional unit, packaged in accordance with 5.1, shall be examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

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4.6.5 Failure of sample. Should any one sample fail to meet any of the specified examinations and test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.8 and 4.9 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever Government designs are not specified (see 4.7.2.1). Since tolerance of test equipment is normally considered to be within ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test

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methods and procedures as specified in 4.8.

TABLE IV. Test Equipment List. 1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1.
DC Voltmeter/Multimeter	Standard - capable of measuring voltages stated in 4.8.1.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.1.
Digital Oscilloscope	HP 54100D or Equivalent - capable of measuring signals described in 4.8.1.4.

NOTE:
1/ Must be calibrated to the requirements of MIL-STD-45662 prior to use.

4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

4.8.1 Performance tests. The following tests and examinations shall be performed to demonstrate compliance with 3.4.2. The PWBA shall be connected to the circuit specified in Figure 6. Apply power by turning ON S13, S14 and S15 and leaving on until all tests have been completed. All other switches shall be as shown in Figure 6 prior to each test specified in 4.8.1.1 through 4.8.1.13.

4.8.1.1 Current drain. Verify that the quiescent current drain at all three ammeters meets the requirements of 3.4.2.1.

4.8.1.2 Stable state. Verify the stable state requirements of 3.4.2.2 by setting S9 to positions 1 through 6 and measuring the voltages.

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4.8.1.3 Wire cut. To verify the wire cut requirements of 3.4.2.3, turn ON S6, set S9 to positions 1 through 6 and measure the voltages.

4.8.1.4 TOW trigger. To verify the TOW trigger requirements of 3.4.2.4, turn ON S6, set S7 to position 1, set S9 to positions 1 through 4 and 6 and measure the voltages. Set S9 to position 5, set S16 to position 2, and check the signal on the digital oscilloscope. With S7 set to position 2 and S6 turned OFF, turn ON S6 and measure the voltages with S9 in positions 1 through 4 and 6 on the voltmeter and check the signal with S9 in position 5 on the digital oscilloscope.

4.8.1.5 COAX trigger. To verify the COAX trigger requirements of 3.4.2.5, set S5 to position 2, set S9 to positions 1 through 6 and measure the voltages.

4.8.1.6 25 mm trigger. To verify the 25mm trigger requirements of 3.4.2.6, turn ON S3, set S5 to position 3, turn ON S8, set S4 to position 2, momentarily turn ON S2 and measure the voltage with S9 at position 1. Set S9 at positions 2 through 6, and for each, turn ON S2 momentarily and measure the voltage. Repeat the above steps for positions 3 through 9 of S4.

4.8.1.7 Weapons disable. To verify the weapons disable requirements of 3.4.2.7, turn ON S3, turn ON S8, set S4 to position 2, momentarily turn ON S2 and measure the voltage with S9 at position 1. Set S9 to positions 2 through 6 and for each, turn ON S2 momentarily and measure the voltage.

4.8.1.8 25 mm trigger disable. To verify the 25mm trigger disable requirements of 3.4.2.8, turn ON S3, set S5 to position 3, set S4 to position 2, momentarily turn ON S2 for positions 1 through 6 of S9 and measure the voltage for each.

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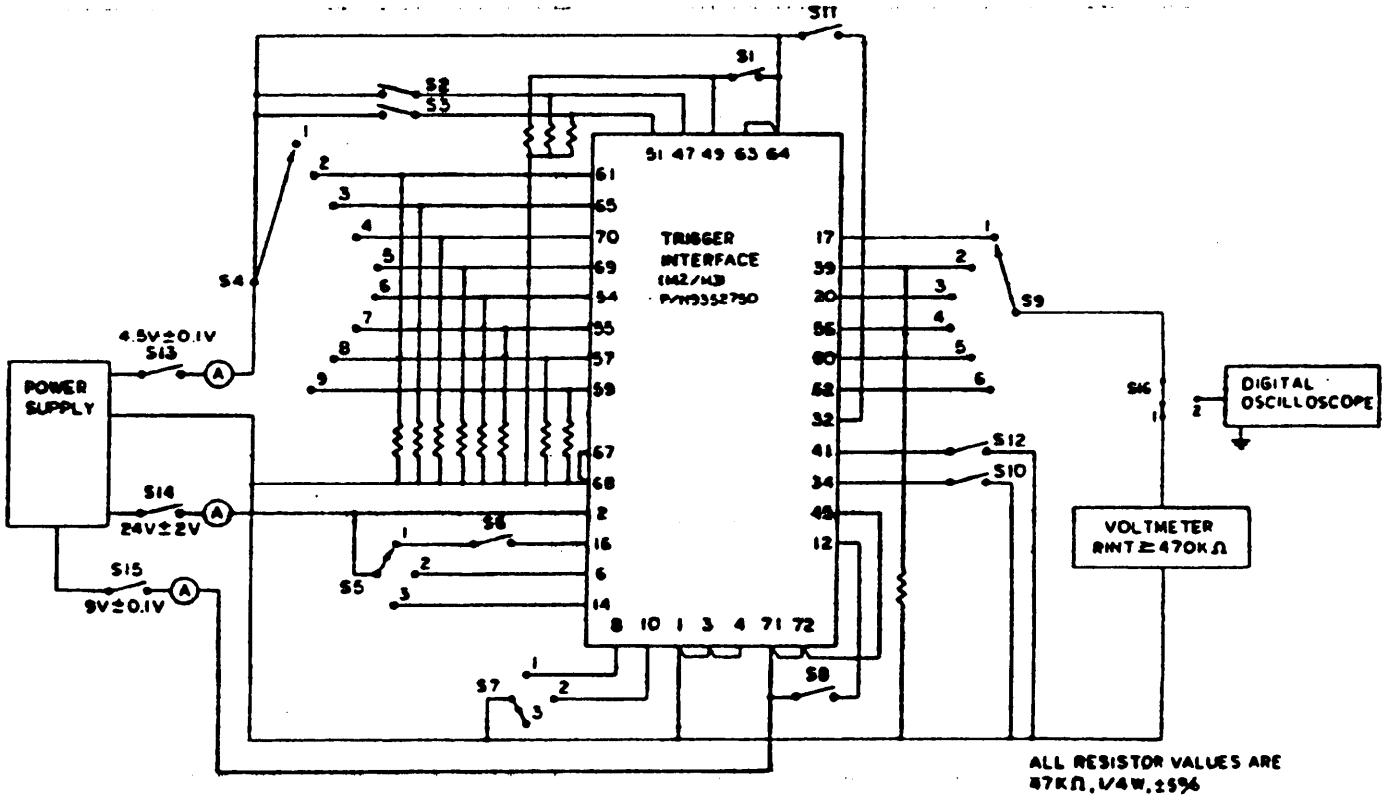


FIGURE 6. Test Circuit for Trigger Interface.

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4.8.1.9 Microphone trigger. To verify the microphone trigger requirements of 3.4.2.9, turn ON S3, set S5 to position 3, turn ON S10, set S4 to position 2, and momentarily turn ON S2 for positions 1 through 6 of S9 and measure the voltages.

4.8.1.10 AWESS trigger. To verify the AWESS trigger requirements of 3.4.2.10, turn ON S11, S3 and S8, set S4 to position 2, set S5 to position 3, and momentarily turn ON S2 for positions 1 through 6 of S9 and measure the voltages.

4.8.1.11 AWESS/25 mm trigger. To verify the AWESS/25mm trigger requirements of 3.4.2.11, turn ON S11, S3, S8 and S10, set S4 to position 2, set S5 to position 3, and momentarily turn ON S2 for positions 1 through 6 of S9 and measure the voltages.

4.8.1.12 Trigger disable. To verify the trigger disable requirements of 3.4.2.12, set S5 to position 3, turn ON S1, S8 and S11 and measure the voltages for positions 1 through 6 of S9.

4.8.1.13 FlashWESS trigger. To verify the flashWESS trigger requirements of 3.4.2.13, turn ON S3, S8, and S12, set S4 to position 2, set S5 to position 3, and momentarily turn ON S2 for positions 1 through 6 of S9 and measure voltages.

4.9 Environmental tests.

a. Unless otherwise specified, environmental tests shall be conducted in accordance with the test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the tests required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.

4.9.1.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501 using the following procedure:

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a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

b. Step 2 - Raise the internal temperature to 62.8°C (145°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 4 hours while insuring the relative humidity is not in excess of 15 percent.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.1.

4.9.2 Low temperature. This satisfies the storage and operating conditions of low temperature requirements as specified in 3.4.3.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502 using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain the temperature for period of 4 hours.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.1.

e. Step 5 - Return the test item, nonoperating, to standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage, and in the absence of physical damage, perform the tests

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required in 4.8.1.

4.9.3 Shock. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 7), shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verification shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.9.4 Vibration. The assembly, when mounted on a fixture that simulates its next higher assembly (see Figure 7), shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes per axis with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.1.

4.10 Certifications. Certification shall be provided that all units have been produced in accordance with the requirements of 3.5 and packaged in accordance with the requirements of 5.1.

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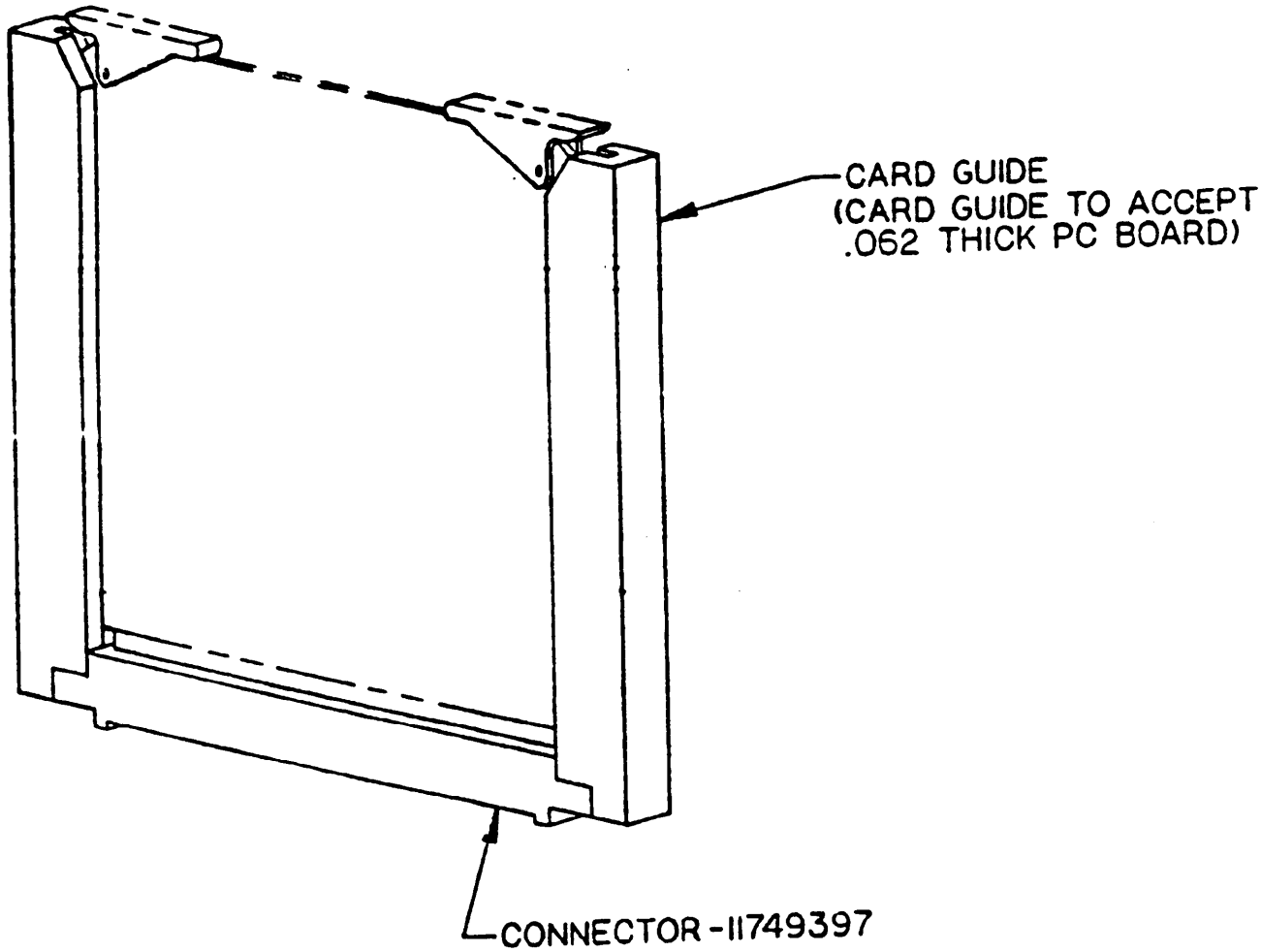


FIGURE 7. Mounting Configuration.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI-P9352750. The level of protection shall be A, B, or C, as defined by the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Trigger Interface PWBA is part of the Multiple Integrated Laser Engagement System.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

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6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

6.4 Definitions.

6.4.1 MILES. Multiple Integrated Laser Engagement System.

6.4.2 Round. A laser message that simulates the effects of a round from the weapon.

6.4.3 SME. Standard Measuring Equipment.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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6.7 Subject term (key word) listing.

Fire Control
Multiple Integrated Laser Engagement System
Training Devices

Custodians:
Army - AR

Preparing activity:
Army - AR

(Project 1265-A033)

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INSTRUCTIONS

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DOCUMENT IMPROVEMENT PROPOSAL	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70875 (AR)	10 MARCH 1992

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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